



MY PROJECT

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BACKGROUND

- Final year **Individual Project**
- ‘Detection of Hate Speech towards the LGBTQIA+ community using lexicon-based sentiment analysis’
- **2 semesters**
- **Report**

OBJECTIVES

- **Develop** a hate speech detection program
- **Positive results**
- **Report**

RESPONSIBILITIES

- Individual project
 - **Just me**
- Decide on the **method** of approach
 - **Research**
- **Gather** any required **data**
 - **Research**
 - **Cleaning data**
- **Program** the solution
- Write a **report**

HATE SPEECH DETECTION - FEASIBILITY

- **Resources**

- Availability of data
 - Third-party
 - I source it

- **Knowledge**

- What is in my scope

- **Time**

- I have a **limited time**
- This will affect **resources** and **knowledge**

HATE SPEECH DETECTION - APPROACHES

- **Machine Learning**

- **Pattern recognition** to classify text as hate speech towards the LGBTQIA+ community or not

- **Lexicon-based Analysis**

- Break the text into the **individual words**
- **Score each word** separately

MACHINE LEARNING

- **Pros:**

- Can be **very accurate**

- **Cons:**

- Requires **training data** (Resource issue)
- **Hard to explain** why the decision was made
- **Complex** to develop (Knowledge issue)

LEXICON-BASED ANALYSIS

- **Pros:**

- Can be **accurate**
- **Easy to explain** why a decision was made

- **Cons:**

- Requires a **dataset** of **values** for words (Resource issue)
- **No context** of words in the overall text

HATE SPEECH DETECTION APPROACH

- Both have benefits and negatives
- **Lexicon-based approach will be used**
 - **Explainability is important** due to hypothetical future applications
 - The removal of context for words has been **proven to work**
 - I am confident that I can **implement** the program in the **time required**

TECHNOLOGIES

- Develop in **Python**
 - Previous experience
 - Popular, so others can read it
- Storage of text was in **.csv**
 - Simple
 - Each record does not need much information

PLAN

- **Text parser**
 - Reformat text, ready for processing
- **Reformat SentiWordNet**
 - This is a database for determining if a word is positive or negative in nature
 - Reformatting to make it as efficient as possible when reading
- **Hate speech detection**
 - Split the text into words
 - Find the corresponding value for each word
 - Determine if the text is hate speech or not

PLAN

- **LGBTQIA+ related speech detection**
 - Cross references words with LGBTQIA+ related words
 - Determines if the text is LGBTQIA+ related or not
- **The controller**
 - The controller manages the other modules
- **Testing**
 - Acquire test data

CHALLENGES AND SKILLS LEARNT

Challenges:

- Finding **reliable third-party data**
 - Positive/negative values for words
 - LGBTQIA+ related words
 - Training data

Skills Learnt:

- **My largest individual project**
 - **Time management** and **planning skills** were used a lot
 - **Program designing** e.g. **class diagrams** and **UML component diagrams**

REFLECTION

- The **results** were **mixed**
- Testing against LGBTQIA+ hate speech
 - **Recall = 0.49**, half of all LGBTQIA+ hate speech was identified
 - **Precision = 0.92, when classified** as LGBTQIA+ hate speech, it would **likely be right**
- What would I do differently?
 - **Machine learning** approach
 - I **started** the project **before** many of my **AI-related modules**
 - I am a lot more **confident** in the topic

